

EXERCISE 07: The Pseudo Inverse, the Adaline and the Bayes rule

Consider a student perceptron with $N = 20$ binary input elements and one binary output receiving from an identical teacher perceptron the classification results for $P = \alpha N$ random examples $\{\vec{\xi}^\alpha\}$ of a training set.

- Apply the Pseudo Inverse rule to evaluate $\vec{J}^{PI}$.
- Repeat the procedure 1000 times in correspondence of different training sets and different teacher perceptrons and evaluate as a function of $\alpha \in (0, 1)$ the average training error and the average test error on a test set with $P_{test} = 1000$.
- What happens when $\alpha \rightarrow 1$?
- Apply now the Adaline rule to evaluate $\vec{J}^A$ using the following steps
 - Initialise a student perceptron as sum of all the examples $\{\vec{\xi}^\mu\}$ in the training set and normalise it to unitary norm.
 - Apply the Adaline correction $\delta\vec{J} = -\gamma \sum_\mu \xi^\mu (\vec{J}\vec{\xi}^\mu - \sigma_T^\mu)$ with $\gamma = 0.05$ iteratively until the cost function $E(J) = 0.5 \sum_\mu (\vec{J}\vec{\xi}^\mu - \sigma_T^\mu)^2$ changes less than 10^{-8} in a single step.
- Repeat the procedure 1000 times in correspondence of different training sets and different teacher perceptrons and evaluate as a function of $\alpha \in (0, 10]$ (with particular attention to the range $\alpha \in (0, 2]$) the average training error and the average test error on a test set with $P_{test} = 1000$.
- Check that for $\alpha < 1$ it superimposes to the result of the Pseudo Inverse rule. What happens for $\alpha > 1$?
- Apply the Bayes rule to evaluate $\vec{J}^B$ as the center of mass of $M = 100$ student perceptrons obtained from the randomised perceptron rule used in Exercise 03.
- Repeat the procedure 1000 times in correspondence of different training sets and different teacher perceptrons and evaluate as a function of $\alpha \in [0, 10]$ the average training error and the average test error on a test set with $P_{test} = 1000$.
- Superimpose the analytic predictions for the Bayes generalisation error ϵ_B obtained as solution of

$$\frac{R_B^2}{\sqrt{1 - R_B^2}} = \frac{\alpha}{\pi} \int Dt \exp\left(-\frac{R_B^2 t^2}{2}\right) / H(-R_B t)$$

with $Dt = \exp(-t^2/2)dt/\sqrt{2\pi}$ and using $\epsilon = \frac{1}{\pi} \arccos(R_B)$.

- Compare them with the results of all other learning rules studied so far with particular attention to the small and large α behaviour.